

Gap Analysis Report – EdTech Study Platform vs. SRS v7.3

Executive Summary

The platform is substantially built. The core learning path, exam engine, auth flows, B2 storage, admin panel, and all seven SRS gating steps are implemented and working. The largest remaining risks are **infrastructure credentials** (nothing storage/email-related actually works in production without them) and a handful of missing features in the auth, admin analytics, and profile layers.

What Is Confirmed Implemented

Before the gaps, to avoid duplicate work:

Area	Status
All 7 SRS gating steps (Lecture→Quiz→DPP→PYQ→TopicTest→ChapterTest→PDFUpload)	 Complete, server-enforced twice
Exam engine (timer, pause/resume with limit, auto-submit sweep, answer grid)	 Complete
QR codes (auto-generated from videoUrl, stored as SVG, shown in results)	 Complete
Auth (register, login, verify-email, resend, forgot-password, reset-password, change-password)	 Complete
B2 operations (upload URL, download URL, delete – profile photos and PDFs)	 Code complete,

Area	Status
	needs secrets
Admin (user approval, roles, progress reset, GDPR export, system config, audit log)	 Complete
Admin question bank (CRUD, CSV bulk import with preview, video solution + QR)	 Complete
Storage monitor (6hr cron, daily midnight sweep, admin dashboard widget)	 Complete
Pomodoro (session log, streak tracking, topicId context, push notification on milestones)	 Complete
Study planner (auto-generate, drag-and-drop reorder, weak-topic priority)	 Complete
PWA manifest + service worker (cache-first, push event handler, background sync)	 Complete
Leaderboard, external test tracker, QR scan history, dashboard	 Complete

CRITICAL Gaps – Production is Broken Without These

C-01 – Missing Environment Secrets (Infrastructure Dead Zone)

Impact: Every B2 operation (profile photos, PDF notes uploads/downloads) silently fails. Every transactional email (verification, welcome, password reset) is never actually sent. Users can register but never verify their email, meaning they can never log in past the verification wall.

Missing secrets:

Variable	Purpose	Effect if Missing
B2_APPLICATION_KEY_ID	Backblaze B2 auth	All file ops return 403
B2_APPLICATION_KEY	Backblaze B2 auth	All file ops return 403
B2_BUCKET_NAME	Target bucket	Upload/download URLs are malformed
RESEND_API_KEY	Transactional email	All emails silently swallowed

Action: Use the Replit Secrets panel to add all four. For B2: create an application key in the Backblaze console scoped to your bucket. For Resend: generate an API key at resend.com and verify your sending domain.

Files affected:

- artifacts/api-server/src/lib/b2.ts – reads B2_APPLICATION_KEY_ID, B2_APPLICATION_KEY, B2_BUCKET_NAME
- artifacts/api-server/src/lib/email.ts – reads RESEND_API_KEY

C-02 – Email Verification is Gated on Sending, But Sending Silently Fails

Impact: Because RESEND_API_KEY is missing, sendVerificationEmail() throws or returns an error, but auth.ts registration continues anyway and responds 201. The user has an account with emailVerified: false and no token in their inbox. They cannot ever log in.

Action: After adding RESEND_API_KEY (C-01), audit the error handling in POST /auth/register to confirm that a Resend failure rolls back the

user row or at minimum returns a clear error. Currently the response is 201 regardless.

Files to check/fix:

- artifacts/api-server/src/routes/auth.ts – register handler, wrap email send in try/catch and handle the failure case
 - artifacts/api-server/src/lib/email.ts – confirm error propagation
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C-03 – Profile Photo Endpoint Has Inconsistent Auth Middleware

Impact: PATCH /api/profile uses requireAuth (pending users can update name/mobile). But DELETE /api/profile/photo uses requireApproved. A pending_approval user who uploaded a photo during a grace window cannot remove it. This is a logic inconsistency that will surface as a silent 403.

Files to fix:

- artifacts/api-server/src/routes/profile.ts – change DELETE /api/profile/photo to use requireAuth to match PATCH, or intentionally document the difference
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HIGH Priority Gaps – Major SRS Features Missing

H-01 – No Email Change Flow (SRS v7.3 Profile Management)

What's missing: The SRS v7.3 explicitly adds "Profile Management" enhancements. The profile.tsx page displays the email address read-only. There is no mechanism to initiate an email change, which would require: (1) the user submits a new email, (2) a verification token is sent to the new address, (3) upon verification the email is swapped.

Files to create/modify:

- artifacts/api-server/src/routes/auth.ts – add POST /auth/request-email-change (generate token, send to new address) and POST /auth/confirm-email-change (verify token, update email)
 - artifacts/api-server/src/lib/email.ts – add sendEmailChangeVerification() template
 - artifacts/edtech-platform/src/pages/profile.tsx – add "Change Email" section with input + "Send verification" CTA
 - lib/api-spec/openapi.yaml – add both endpoints then run codegen
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H-02 – Mobile OTP Verification Not Implemented (SRS v7.3 §Enhanced Registration)

What's missing: SRS v7.3 title explicitly says "Mandatory Indian Mobile Validation." The current implementation validates the +91 format via regex, which satisfies format correctness, but there is zero OTP delivery or confirmation. A user can register with any syntactically valid number they don't own.

Clarification note: If the intent is format-only (no SMS provider), the current implementation is correct and this gap can be formally accepted. If real OTP is required, you need an SMS gateway (e.g. MSG91, Twilio, or Fast2SMS for India).

Files to create/modify (if OTP is required):

- artifacts/api-server/src/routes/auth.ts – add POST /auth/send-mobile-otp and POST /auth/verify-mobile-otp
- artifacts/api-server/src/lib/sms.ts – new file, SMS provider client
- lib/db/src/schema/users.ts – add mobileVerified, mobileOtpToken, mobileOtpExpiry columns
- artifacts/edtech-platform/src/pages/register.tsx – add OTP step after form submission
- pnpm --filter @workspace/db run push – after schema change

H-03 – No Admin QR Scan Analytics Dashboard

What's missing: QR scans are logged to qr_scans table via qr-scans.ts, but there is no admin UI showing which questions have the most video solution scans, which students are using QR codes, or scan trends over time. The data is being collected but never surfaced.

Files to create/modify:

- artifacts/api-server/src/routes/admin.ts – add GET /admin/qr-analytics returning per-question scan counts and recent scan list
 - artifacts/edtech-platform/src/pages/admin-dashboard.tsx – add a "Video Solution Engagement" section below storage monitor
 - lib/api-spec/openapi.yaml + codegen run
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H-04 – No Live Exam Session Monitor for Admins

What's missing: There is no admin view to see currently in-progress exam attempts (students who are live in an exam right now). The auto-submit sweep handles expiry, but admins have no visibility into who is mid-exam, their time remaining, or if an attempt is stuck.

Files to create/modify:

- artifacts/api-server/src/routes/admin.ts – add GET /admin/live-attempts (query examAttemptsTable where status = 'in_progress', join with user and exam name)
 - artifacts/edtech-platform/src/pages/admin-dashboard.tsx – add a "Live Exams" widget (auto-refreshing every 30s)
 - lib/api-spec/openapi.yaml + codegen
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H-05 – Subject/Chapter/Topic Bulk Import Missing

What's missing: The question bank has a full CSV import UI with validation preview. But creating the content hierarchy (Subjects → Chapters → Topics) is one-at-a-time only. For an admin seeding a full JEE/NEET syllabus with hundreds of topics, this is a critical productivity gap.

Files to create/modify:

- artifacts/api-server/src/routes/admin.ts or new artifacts/api-server/src/routes/content-import.ts – add POST /admin/import-syllabus accepting CSV with columns: subject_name, chapter_name, topic_name, topic_order
 - artifacts/edtech-platform/src/pages/admin-subjects.tsx – add an "Import Syllabus" button with the same CSV preview UI pattern already used in admin-questions
 - lib/api-spec/openapi.yaml + codegen
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H-06 – Telegram Lecture Deep Links Not Integrated

What's missing: The SRS §6 specifies "Telegram Lecture Delivery: Deep link integration for video content with click tracking." The current system tracks lectureClickCount in topic_progress when a student clicks the lecture, but there is no telegramUrl or lectureUrl field on topics, and the topic detail page has no lecture viewer or link button.

Files to create/modify:

- lib/db/src/schema/topics.ts – add telegramUrl text and youtubeUrl text columns
- pnpm --filter @workspace/db run push
- artifacts/api-server/src/routes/topics.ts – accept telegramUrl/youtubeUrl in create/update, expose in GET

- artifacts/edtech-platform/src/pages/topic-detail.tsx – render a "Watch Lecture" button that opens the link and calls POST /topics/:topicId/lecture-click
 - lib/api-spec/openapi.yaml + codegen
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MEDIUM Priority Gaps – UX and Completeness

M-01 – No Grand Test / Subject-Level Test Gating UI Exposure

What's implemented: The backend gates Subject Tests behind all Chapter Tests being passed. But the topic/subject detail pages may not visibly show students when a Subject Test or Grand Test unlocks and where to access it.

Action: Audit artifacts/edtech-platform/src/pages/subject-detail.tsx and chapter-detail.tsx. Confirm there is a "Subject Test" CTA card that appears once all chapter tests are done, and a "Grand Test" card at the dashboard level.

M-02 – No Account Self-Deactivation / Data Deletion for Users

What's missing: GDPR-style: admins can export user data, but users themselves have no way to request account deletion from their profile page.

Files to modify:

- artifacts/api-server/src/routes/profile.ts – add DELETE /api/profile (mark user deleted, anonymize PII, schedule B2 cleanup)

- `artifacts/edtech-platform/src/pages/profile.tsx` – add a "Delete Account" section with confirmation dialog
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M-03 – Storage Management Is Read-Only (Admin Cannot Purge Orphans)

What's missing: The storage monitor shows GB used and alerts at threshold. But an admin cannot identify or delete orphaned B2 files (files in B2 that have no corresponding DB record, left over from interrupted uploads).

Files to modify:

- `artifacts/api-server/src/lib/storage-monitor.ts` – add an orphan detection function comparing B2 bucket listing against `notesTable + usersTable.photoB2Key`
 - `artifacts/api-server/src/routes/admin.ts` – add GET `/admin/storage/orphans` and DELETE `/admin/storage/orphans`
 - `artifacts/edtech-platform/src/pages/admin-dashboard.tsx` – add orphan list widget with bulk-delete button
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M-04 – No "Remember Me" / Extended Session

What's missing: JWTs are fixed-expiry. There is no `rememberMe` toggle that issues a longer-lived token or a refresh token. Students studying daily get logged out unexpectedly.

Files to modify:

- `artifacts/api-server/src/routes/auth.ts` – accept `rememberMe: boolean` in login, issue 30d JWT vs 24h
- `artifacts/edtech-platform/src/pages/login.tsx` – add "Remember me" checkbox

M-05 – Push Notification Subscribe Endpoint Does Not Persist Subscriptions

What's implemented: POST /notifications/subscribe accepts a push subscription object and returns { success: true } – but **does not save it to the database**. Push notifications sent by the server will never reach the user's browser because there is no stored endpoint to push to.

Files to modify:

- lib/db/src/schema/ – create push_subscriptions.ts table (userId, endpoint, p256dh, auth, createdAt)
- lib/db/src/schema/index.ts – barrel export
- pnpm --filter @workspace/db run push
- artifacts/api-server/src/routes/notifications.ts – persist subscription in subscribe, add DELETE /notifications/subscribe for unsubscribe
- artifacts/api-server/src/routes/pomodoro.ts / exams.ts – use stored subscriptions to send actual Web Push payloads (requires web-push npm package + VAPID keys)
- Add VAPID_PUBLIC_KEY and VAPID_PRIVATE_KEY to secrets

M-06 – No Cloudflare CDN in Front of B2 (Cost Risk)

What's missing: The SRS specifically requires Cloudflare in front of B2 to minimize egress costs (B2 free egress only applies when served via Cloudflare). Without it, every PDF download and profile photo view costs egress fees after the free allowance.

Action: This is infrastructure configuration, not code. Steps:

1. Connect your B2 bucket to a Cloudflare-proxied custom domain (e.g. cdn.yourdomain.com → f000.backblazeb2.com)

2. Add B2_CDN_BASE_URL secret
3. In artifacts/api-server/src/lib/b2.ts, use CDN URL for getDownloadSignedUrl responses (public files) and keep presigned for private files

M-07 – Weak Topic Detection in Auto-Planner Uses Stub Logic

Action: Audit artifacts/api-server/src/routes/tasks.ts – specifically the auto-generate function. Confirm it actually queries exam_results to identify topics where the student's average score is below the passing threshold, not just returning all uncompleted topics.

Low Priority / Future Enhancements

ID	Gap	Notes
L-01	OAuth / Social Login (Google)	No provider configured; requires OAuth app setup
L-02	TOTP / MFA	No authenticator app support
L-03	Session management UI (view/revoke active sessions)	Requires storing JWT jti in DB
L-04	Password-less login (magic links)	Different email template + token flow
L-05	Question item analysis (high-skip/always-wrong flagging)	Requires aggregating examAnswersTable
L-06	Sentry error monitoring integration	SENTRY_DSN not configured anywhere

ID	Gap	Notes
L-07	Offline exam answering via IndexedDB + background sync	sw.js has sync stub, Dexie.js not installed
L-08	PDF annotation with pdf-lib	Notes viewer is render-only, no annotation layer

Implementation Roadmap (Logical Order)

Phase 1 – Infrastructure (Unblocks Everything Else)

1. Add B2_APPLICATION_KEY_ID, B2_APPLICATION_KEY, B2_BUCKET_NAME to Replit Secrets
2. Add RESEND_API_KEY to Replit Secrets
3. Fix auth.ts registration to handle email send failure gracefully (C-02)
4. Fix profile.ts photo-delete auth middleware (C-03)
5. Set up Cloudflare in front of B2 bucket, add B2_CDN_BASE_URL (M-06)

Phase 2 – Critical Feature Completion

6. Persist push subscriptions to DB + add web-push package + VAPID keys (M-05)
7. Add Telegram/YouTube URL fields to topics + lecture click tracking UI (H-06)
8. Add email change flow – backend endpoints + profile UI (H-01)

Phase 3 – Admin Completeness

9. Add live exam session monitor widget in admin dashboard (H-04)
10. Add QR scan analytics dashboard (H-03)
11. Add syllabus bulk import (CSV: subject/chapter/topic) (H-05)
12. Add orphaned B2 file detection + purge UI (M-03)

Phase 4 – UX Polish

13. Add "Remember Me" to login (M-04)
14. Add account self-deletion from profile (M-02)
15. Audit and confirm Grand Test unlock CTA on subject-detail page (M-01)
16. Audit auto-planner weak topic detection query (M-07)

Phase 5 – Decision Point

17. Decide: formal mobile OTP (needs SMS provider) or document format-only validation as accepted (H-02)
18. Add Sentry DSN and initialize in both api-server and frontend (L-06)

Quick Reference: Files That Need the Most Attention

Priority	File	What's Needed
 Critical	artifacts/api-server/src/routes/auth.ts	Handle email send failure on register
 Critical	artifacts/api-server/src/routes/profile.ts	Fix DELETE /photo auth middleware

Priority	File	What's Needed
 Critical	Replit Secrets panel	Add 4 missing secrets
 High	artifacts/api-server/src/routes/notifications.ts	Persist push subscriptions to DB
 High	lib/db/src/schema/topics.ts	Add telegramUrl, youtubeUrl columns
 High	artifacts/api-server/src/routes/auth.ts	Add email-change endpoints
 High	artifacts/api-server/src/routes/admin.ts	Add live-attempts + QR analytics + syllabus import
 Medium	artifacts/api-server/src/lib/b2.ts	Support CDN base URL for downloads
 Medium	artifacts/edtech-platform/src/pages/profile.tsx	Email change UI + account deletion
 Medium	artifacts/edtech-platform/src/pages/login.ts	